

DEBJIT DAS

B.Tech – Electronics & Communication Engineering (Lateral Entry) | Diploma – Electrical Engineering

Jalpaiguri Government Engineering College • 3rd Year, 6th Semester

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CAREER OBJECTIVE

Final-year B.Tech ECE student with a Diploma in Electrical Engineering and hands-on experience across the full electromechanical stack — from power electronics and electrical machines to embedded firmware, sensor fusion, and edge AI. Seeking a placement role in embedded systems, industrial automation, or intelligent hardware where deep domain knowledge of both electrical and communication engineering creates genuine engineering value. Demonstrated ability to build production-relevant systems — most recently an AI-controlled robotic hand achieving clinical-grade sEMG control at 34× lower cost than commercial alternatives.

EDUCATION

Jalpaiguri Government Engineering College (JGEC) B.Tech in Electronics & Communication Engineering — Lateral Entry • 3rd Year, 6th Semester	Jan 2024 – Jun 2027
Central Calcutta Polytechnic (Govt. of West Bengal) Diploma in Electrical Engineering — 84% • 1st Class with Distinction	Nov 2021 – Aug 2024
Chatra Nandalal Institution Madhyamik — WBBSE • 87%	2015 – 2021

3rd Sem GPA	4th Sem GPA	5th Sem GPA	CGPA (up to 5th Sem)	Equivalent %
7.57	8.29	7.17	7.68	76.8%

PROJECTS

AI-Based Robotic Hand Control Using sEMG & Edge AI ESP32 • MyoWare 2.0 sEMG • MediaPipe • TinyML CNN • PCA9685 • MG946R Servos • FlexiForce • MPU-6050 • ESP-NOW • Embedded C • Python	B.Tech Major Project • Ongoing
- Designed and built a full-stack AI-controlled robotic hand combining Bayesian sensor fusion of MediaPipe hand-landmark vision (30 FPS, 21 landmarks) and MyoWare 2.0 sEMG forearm signals — reducing false positives by 60% vs. single-sensor systems and enabling robust occlusion recovery. - Deployed a TinyML CNN on ESP32 for autonomous slip detection with <50 ms latency; system autonomously corrects grip without user intervention — validated across foam ball, plastic bottle, and rigid object gripping scenarios. - Implemented a 5-finger independent vibrotactile haptic feedback wristband using ESP-NOW peer-to-peer protocol (~2 ms end-to-end latency, no router required) — blindfolded operator achieved >70% object-density identification accuracy over 20 trials. - Engineered complete system at Rs. 14,604 total BOM cost — 34× cheaper than commercial alternatives (Rs. 1–5 lakh) — using InMoov open-source 3D-printed hand, 6× MG946R servos driven by PCA9685 12-bit PWM, and custom buck-regulated 11.1V Li-Ion power architecture.	

Automatic Obstacle Avoidance Car Arduino Uno • L293D Motor Driver • HC-SR04 Ultrasonic Sensor • Servo Motor • Embedded C	Diploma Major Project • 2023–24
- Built an autonomous four-directional vehicle using HC-SR04 ultrasonic ranging, L293D H-bridge motor control, and real-time servo-based obstacle scanning — fully programmed in Embedded C on Arduino Uno. - Demonstrated hardware-software co-design, microcontroller peripheral interfacing, and closed-loop real-time control — forming the practical embedded foundation extended in the B.Tech robotic hand project.	

INTERNSHIPS

Telecom Technology Intern — Bharat Sanchar Nigam Limited (BSNL) Kalyani, West Bengal • Hybrid Mode • Certificate Awarded	Jun – Jul 2025 • 2 Months
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- Worked at a live BSNL exchange facility — examined DSLAM equipment configuration, OFC splicing infrastructure, and IP-based broadband network architecture supporting active subscriber connections in Kalyani district.
- Developed practical understanding of last-mile broadband delivery, SDH transmission hierarchy, and telecom network operations — directly applicable to 5G NR infrastructure and communication systems roles.

AutoCAD Engineering Intern — Larsen & Toubro (L&T)

May – Jun 2024 • 2 Months

Serampore, West Bengal • On-Site • Certificate Awarded

- Produced construction-grade 2D technical drawings and 3D solid models for electrical distribution equipment at an active L&T industrial facility — drawings validated against design specifications and L&T project documentation standards.
- Created Bill of Materials (BOM) for panel assemblies; cross-verified component specifications against engineering drawings — developing industrial documentation discipline directly applicable to embedded hardware design and system integration roles.

TECHNICAL SKILLS

Embedded & Hardware	AI / Software	Electrical & Power	Telecom & Protocols
ESP32, Arduino Uno Embedded C / C++ PCA9685, L293D Servo & Sensor Integration Hardware-SW Co-design	Python (NumPy, Pandas) MediaPipe, TinyML CNN TensorFlow Lite (ESP32) Signal Processing	AutoCAD 2D & 3D Electrical Machines Power Electronics Buck/Boost Converters Battery Management PLC Basics (Ladder)	I2C, SPI, UART ESP-NOW, OFC DSLAM / IP Networks Analog & Digital Comms CAN Bus (Fundamentals)

AREAS OF INTEREST

Embedded Systems & IoT ♦ Industrial Automation & Robotics ♦ Edge AI / TinyML ♦ Automotive Electronics ♦ Power Electronics & EV Systems ♦ Communication Systems ♦ Semiconductor Devices ♦ VLSI Design Fundamentals ♦ Biomedical Instrumentation

KEY ACADEMIC COURSEWORK

- ♦ Electronic Devices & Circuits
 - ♦ Analog & Digital Communication Systems
 - ♦ Microprocessors & Microcontrollers
 - ♦ VLSI Design Fundamentals
- ♦ Digital Electronics & Logic Design
 - ♦ Signals & Systems
 - ♦ Control Systems Engineering
 - ♦ Network Theory & Transmission Lines

CERTIFICATIONS & ACHIEVEMENTS

- ✓ BSNL Telecom Technology Internship Certificate

Jun – Jul 2025 • Kalyani, West Bengal
- ✓ L&T AutoCAD Internship Certificate (2D & 3D Modeling)

May – Jun 2024 • Serampore, West Bengal
- ✓ Diploma in Electrical Engineering — 1st Class with Distinction (84%)

Central Calcutta Polytechnic, Govt. of West Bengal • 2024
- ✓ Madhyamik (WBBSE) — 87%

Chatra Nandalal Institution • 2021

LANGUAGES

Bengali (Native) | English (Professional Working Proficiency)

I hereby declare that all information furnished above is true, complete, and correct to the best of my knowledge and belief.

Place: Jalpaiguri

Date: May 2026

Debjit Das